
A Strictly Proper Scoring Rule and a Calibration Metric for Interval-Censored Data Analysis

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Abstract

Interval-censored data present unique challenges in statistical analysis due to the partial observability of event times within known intervals, requiring assumptions about the censoring mechanism. This paper explores the theoretical relationship between two foundational assumptions: independent monitoring and non-informative censoring. We demonstrate that these assumptions are equivalent for Case-1 interval-censored data, but not for Case- K interval-censored data, where $K \geq 2$, through a synthetic dataset example. Additionally, we propose the first strictly proper scoring rule and calibration metric specifically designed for interval-censored data under the constant-sum assumption and the non-informative censoring assumption, respectively. Our empirical evaluations on real-world datasets show that a neural network model trained with our scoring rule is competitive with established statistical baselines, offering enhanced flexibility. These contributions provide significant advancements in the theoretical understanding and practical analysis of interval-censored data.

1. Introduction

Interval-censored data are ubiquitous across diverse domains, including medical research, reliability engineering, and the social sciences (Huang & Wellner, 1997; Zhang & Sun, 2010; Klein et al., 2014; Bogaerts et al., 2017). In such settings, the exact timing y of an event of interest remains latent, known only to occur within a specific observed interval. The primary objective of interval-censored data analysis is to estimate a predictive model that outputs a conditional cumulative distribution function (CDF), $F(y|\mathbf{x})$, given a vector of covariates $\mathbf{x}$.

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Because event times are only partially observable, statistical inference necessitates specific assumptions regarding the censoring mechanism. Traditionally, the literature relies on three hierarchical assumptions: (i) independent monitoring, where monitoring times are independent of the event time distribution; (ii) non-informative censoring, a weaker condition stating that the censoring process provides no additional information about the target variable; and (iii) the constant-sum assumption, which represents the most general condition for consistency. In this paper, we critically examine the distinction between assumptions (i) and (ii). We construct a synthetic Case-2 interval-censored dataset that satisfies non-informative censoring while violating independent monitoring, thereby delineating the boundaries of these conditions. Furthermore, we provide a formal proof establishing the equivalence of these two assumptions specifically for Case-1 interval-censored data.

A core contribution of this work is a rigorous theoretical treatment of strictly proper scoring rules for interval-censored data. In probabilistic modeling, the use of strictly proper scoring rules is essential to ensure that the predicted distribution converges to the ground truth (Gneiting & Raftery, 2007). Although Rindt et al. (2022) established such rules for survival analysis, previous interval-censoring scoring rules (Avati et al., 2020) lacked formal proofs of properness. We bridge this gap by introducing two nuanced definitions of properness: strictly Y -proper and strictly LR -proper. We prove that the standard scoring rule is strictly Y -proper under assumption (i) and, significantly, strictly LR -proper under the more general constant-sum assumption (iii). This result refines the theoretical landscape, as it extends the validity of these scoring rules beyond the restrictive independent monitoring assumption required in previous survival analysis frameworks (Rindt et al., 2022).

Furthermore, we propose the first calibration metric specifically designed for interval-censored analysis under the non-informative censoring assumption. Assessing whether a model's predicted probabilities $F(y|\mathbf{x})$ align with empirical frequencies is notoriously difficult when event times are unobservable. While calibration metrics exist for survival analysis (Haider et al., 2020; Yanagisawa, 2023; Choi et al., 2025; Yanagisawa & Akiyama, 2025), a specialized metric

Example for $\mathbf{M} = (4, 7)$ and $\mathbf{E} = (0, 1, 0)$

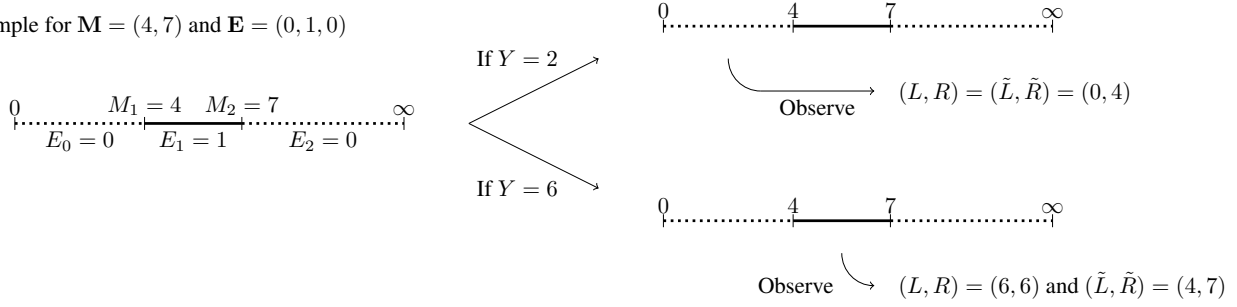


Figure 1. Illustration of the PIC data generation process for $\mathbf{M} = (4, 7)$ and $\mathbf{E} = (0, 1, 0)$. The diagram shows how a latent event time Y is mapped to either an interval or an exact observation depending on the indicator E . If $Y = 2$, $Y \in (M_0, M_1] = (0, 4]$ and $E_0 = 0$, resulting in $L = \tilde{L} = 0$ and $R = \tilde{R} = 4$. If $Y = 6$, $Y \in (M_1, M_2] = (4, 7]$ and $E_1 = 1$, resulting in $L = R = 6$, $\tilde{L} = 4$, and $\tilde{R} = 7$.

for the broader class of interval-censored data has remained elusive. Our proposed metric addresses this void, providing a robust tool for model evaluation.

Finally, we demonstrate the practical utility of our theoretical contributions through empirical evaluations on real-world datasets. By employing our strictly proper scoring rule as a loss function for training neural networks, we show that our approach yields performance that is competitive with, or superior to, established statistical baselines such as ICcforest (Yao et al., 2019) and icenReg (Anderson-Bergman, 2017), while leveraging the high-dimensional flexibility of neural networks.

In summary, this paper advances the theoretical and methodological foundations of interval-censored data analysis by: (1) Formalizing the first strictly proper scoring rule for interval-censored data under censoring assumptions. (2) Developing the first calibration metric suitable for interval-censored data under non-informative censoring.

2. Interval-Censored Data

In many empirical applications, the target variable of interest, Y , is unobservable. Instead of direct observations, we observe a lower bound L and an upper bound R such that $L \leq Y \leq R$, where $L \in \mathbb{R} \cup \{-\infty\}$ and $R \in \mathbb{R} \cup \{\infty\}$ are random variables. Given a covariate vector $\mathbf{X}$, our primary objective is to estimate the conditional cumulative distribution function (CDF):

$$F_Y(y|\mathbf{x}) = \Pr(Y \leq y|\mathbf{X} = \mathbf{x}).$$

Throughout this paper, we assume that $F_Y(y|\mathbf{x})$ is absolutely continuous with density $f_Y(y|\mathbf{x})$.

We consider two fundamental classes of censoring prevalent in the literature: Case- K interval-censored data and Partly Interval-Censored (PIC) data.

2.1. Case- K Interval-Censored Data

Case- K interval-censored data (Huang & Wellner, 1997; Schick & Yu, 2000; Klein et al., 2014) characterize settings where subjects are monitored at K discrete timestamps. Let $\mathbf{M} = (M_1, M_2, \dots, M_K)$ denote the vector of monitoring times such that $0 = M_0 < M_1 < \dots < M_K < M_{K+1} = \infty$. The generative process involves drawing $Y \sim F_Y$ and $\mathbf{M} \sim F_{\mathbf{M}}$, where $F_{\mathbf{M}}$ is the joint distribution of monitoring times. The observer records the interval $(L, R] = (M_i, M_{i+1}]$ that satisfies $M_i < Y \leq M_{i+1}$. Thus, the observed data consists of the narrowest interval defined by the monitoring schedule that contains Y . Consistent with standard treatments (Ng, 2002), we define these as half-open intervals to resolve ambiguities regarding boundary values.

Case- K censoring is typical in clinical longitudinal studies, such as HIV onset detection (Zhang & Sun, 2010), where infection status is only known to change between two successive clinical visits.

Remark. While some literature suggests that Case- K interval-censored data ($K > 2$) can be reduced to Case-2 interval-censored data (Huang & Wellner, 1997), this reformulation often violates the underlying distributional assumptions (see Sec. 3) of the data-generating process (Schick & Yu, 2000). Consequently, we treat Case-2 and Case- K as distinct categories.

2.2. Partly Interval-Censored (PIC) Data

Partly Interval-Censored (PIC) data (Huang, 1999; Kim et al., 2024; Choi et al., 2025) generalize the Case- K framework by permitting the inclusion of exact observations (i.e., $L = R = Y$). Unlike Case- K data, which assumes a fixed number of monitoring occasions K , the PIC framework accommodates a variable number of monitoring times. We define the monitoring schedule as a random vector $\mathbf{M} = (M_1, M_2, \dots, M_{|\mathbf{M}|})$ of stochastic length $|\mathbf{M}|$.

To formalize the generative process, we introduce an auxil-

iary binary vector $\mathbf{E} = (E_0, E_1, \dots, E_{|\mathbf{M}|}) \in \{0, 1\}^{|\mathbf{M}|+1}$, where each E_i serves as an indicator for whether an event occurring within the interval $(M_i, M_{i+1}]$ is exactly observed. For a latent event time $Y \in (M_i, M_{i+1}]$, the resulting observation (L, R) is determined as follows:

- If $E_i = 0$, the event is interval-censored: $L = M_i$ and $R = M_{i+1}$.
- If $E_i = 1$, the event is exactly observed: $L = R = Y$.

In both scenarios, we define the underlying monitoring boundaries as $\tilde{L} = M_i$ and $\tilde{R} = M_{i+1}$. This process is illustrated in Figure 1. Generally, an observation is classified as *left-censored* if $L = 0$, *right-censored* if $R = \infty$, *exact* if $L = R$, and *interval-censored* otherwise.

The PIC framework provides a unified structure for diverse censoring schemes. Specifically, standard survival analysis (Klein et al., 2014; Wang et al., 2019), typically comprising only right-censored and exact data, can be represented by $|\mathbf{M}| = 1$ and $\mathbf{E} = (1, 0)$, where M_1 denotes the censoring time. Furthermore, the framework encompasses doubly-censored data ($|\mathbf{M}| = 2, \mathbf{E} = (0, 1, 0)$) used in (Wellner & Zhan, 1997; Kim et al., 2024) and mixed interval-censored data (Yu et al., 2001).

2.3. Supports of Random Variables

We assume the support of Y is a continuous half-open interval $D_Y = (a, b]$, where $a \in \mathbb{R} \cup \{-\infty\}$ and $b \in \mathbb{R} \cup \{\infty\}$. Unless otherwise specified, we set $a = 0$ and $b = \infty$. This formulation excludes discontinuous supports (see (Oller & Melis, 2023) for treatments of such cases).

Let D_{LR} denote the union of the supports of L and R . We assume $D_{LR} \subseteq D_Y$ and that D_{LR} is discrete, though our theoretical results naturally extend to continuous monitoring distributions. For PIC data, let D_{exact} be the union of the half-open intervals $(M_i, M_{i+1}]$ such that $E_i = 1$. The total support is defined as $D_{\text{PIC}} = D_{\text{exact}} \cup D_{LR}$. Finally, we define $J = \{(l, r) \mid \Pr(L = l, R = r) > 0\}$ as the joint support of (L, R) , and $J_y = \{(l, r) \in J \mid l < y \leq r\}$ as the set of all possible observed intervals that could contain a specific value y .

3. Assumptions on the Censoring Mechanism

Statistical inference for interval-censored data is fundamentally constrained by the partial observability of the target variable Y . Without formal assumptions regarding the censoring mechanism, the distribution F_Y is non-identifiable. This challenge was pioneered by Tsiatis (1975), who demonstrated the non-identifiability of failure-time distributions in survival analysis—a result that directly extends to the PIC framework. Similarly, Betensky (2000) established analo-

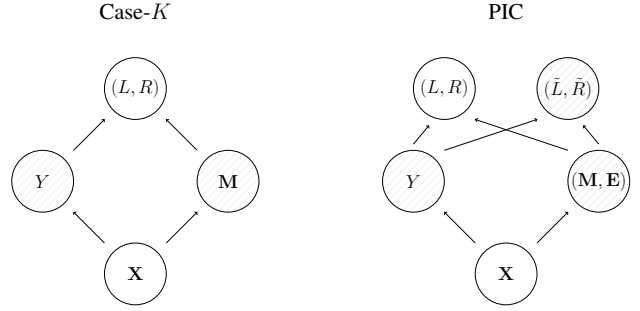


Figure 2. Graphical depiction of the dependency structure among random variables $\mathbf{X}$, Y , $\mathbf{M}$, $\mathbf{E}$, L , R , $\tilde{L}$, and $\tilde{R}$ for Case- K interval-censored and PIC data under the Independent Monitoring assumption (Assumption 3.1), where only $\mathbf{X}$, L , and R are observable.

gous non-identifiability results for Case-1 interval-censored data. To facilitate consistent estimation, we consider three hierarchical assumptions: (i) independent monitoring, (ii) non-informative censoring, and (iii) the constant-sum assumption.

3.1. Independent Monitoring

The most restrictive condition, independent monitoring, postulates that the monitoring schedule $\mathbf{M}$ (and the indicator vector $\mathbf{E}$ in the PIC case) is statistically independent of the event time Y , conditioned on the covariates $\mathbf{X}$. This structure is visually represented in the directed acyclic graphs (DAGs) in Figure 2.

Assumption 3.1 (Independent Monitoring). Conditioned on the covariates $\mathbf{X}$, the event-time variable Y and the monitoring process are independent:

$$Y \perp\!\!\!\perp (\mathbf{M}, \mathbf{E}) \mid \mathbf{X}.$$

Due to its simplicity, this assumption is widely adopted in the existing literature for both Case- K and PIC data analysis (Huang & Wellner, 1997; Anderson-Bergman, 2017; Kim et al., 2024; Choi et al., 2025).

3.2. Non-Informative Censoring

A more flexible alternative is the non-informative censoring assumption. This condition asserts that the observed interval $(L, R]$ conveys no additional information regarding the exact value of Y beyond the constraint that $Y \in (L, R]$ (Zhang & Sun, 2010; Klein et al., 2014; Cho et al., 2022).

Assumption 3.2 (Non-Informative Censoring). For any $(l, r) \in J$, covariates $\mathbf{x}$, and $y \in D_Y$:

$$\begin{aligned} \Pr(Y \leq y \mid L = l, R = r, L < Y \leq R, \mathbf{X} = \mathbf{x}) \\ = \Pr(Y \leq y \mid l < Y \leq r, \mathbf{X} = \mathbf{x}). \end{aligned}$$

Non-informative censoring is strictly weaker than independent monitoring (Lawless, 2004; Piqué, 2006), allowing for certain dependencies between the timing of observations and the event itself, provided the interval boundaries do not “leak” information about the specific location of Y within those boundaries.

3.3. The Constant-Sum Assumption

The most general condition used in this context is the constant-sum assumption (Oller et al., 2004; Piqué, 2006), which serves as a necessary condition for the identifiability of $F_Y(y)$ (Piqué, 2006; Oller et al., 2007).

Assumption 3.3 (Constant-Sum). For any $y \in D_Y$,

$$\sum_{(l,r) \in J_y} k(l, r|\mathbf{x}) = 1,$$

where $k(l, r|\mathbf{x}) = \Pr(L = l, R = r | l < Y \leq r, \mathbf{X} = \mathbf{x})$.

This property generalizes the identifying assumptions for right-censored data (Williams & Lagakos, 1977) and Case-1 interval-censored data (Betensky, 2000).

3.4. Relationship and Practical Implications

Crucially, these assumptions are empirically indistinguishable using only the observable data (L, R) (Lawless, 2004). Consequently, selecting the appropriate framework requires a domain-specific understanding of the data-generating process.

The distinction between independent monitoring (Assumption 3.1) and non-informative censoring (Assumption 3.2) is particularly salient for Case- K ($K \geq 2$) interval-censored data. In Appendix A.1, we provide a synthetic Case-2 dataset that satisfies non-informative censoring while explicitly violating independent monitoring. This highlights that independence is a strong requirement that may not hold in practice, necessitating estimators robust to weaker assumptions.

Conversely, for Case-1 data, we provide a formal proof in Appendix A.2 showing that Assumption 3.2 implies Assumption 3.1, establishing their equivalence in the Case-1 regime.

4. Strictly Proper Scoring Rules

Strictly proper scoring rules (Gneiting & Raftery, 2007) are foundational to both statistical inference and machine learning. Their adoption ensures that predictive models are rigorously evaluated and optimized toward providing calibrated and honest probabilistic estimates.

4.1. Standard Probabilistic Analysis

We briefly review strictly proper scoring rules for fully observable targets (Gneiting & Raftery, 2007). Let $\hat{F}_Y$ be an estimate of the true conditional CDF F_Y .

Definition 4.1 (Proper Scoring Rule). A scoring rule $\mathcal{S}$ is *proper* if:

$$\mathbb{E}_Y[\mathcal{S}(\hat{F}_Y, Y)] \geq \mathbb{E}_Y[\mathcal{S}(F_Y, Y)]. \quad (1)$$

Definition 4.2 (Strictly Proper Scoring Rule). A proper scoring rule $\mathcal{S}$ is *strictly proper* if the equality in (1) holds if and only if $\hat{F}_Y(y) = F_Y(y)$ for all $y \in D_Y$.

A canonical example is the logarithmic score, defined as:

$$\mathcal{S}_{\log}(\hat{F}_Y, y) = -\log \frac{d}{dy} \hat{F}_Y(y). \quad (2)$$

In practice, a scoring rule is informative only when it is strict; a trivial rule that returns a constant for any $\hat{F}_Y$ is proper but lacks the discriminative power necessary for model selection or optimization via empirical risk minimization.

Strictly proper scoring rules are particularly advantageous in the context of machine learning, as they permit the estimation of a conditional cumulative distribution function (CDF), $\hat{F}_Y(y|\mathbf{x})$, through the framework of empirical risk minimization. Given a dataset $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$, the task is reduced to identifying a CDF that minimizes the empirical loss:

$$\frac{1}{N} \sum_{i=1}^N \mathcal{S}(\hat{F}_Y(y^{(i)}|\mathbf{x}^{(i)}), y^{(i)}),$$

where $\mathcal{S}$ denotes a strictly proper scoring rule.

While these scoring rules are primarily utilized to evaluate and optimize covariate-conditional distributions, we follow standard convention in the decision-theoretic literature by omitting the explicit notation of covariates $\mathbf{x}$ throughout the remainder of this section to ensure clarity in the theoretical exposition.

4.2. Extension to Interval-Censored Data

To adapt scoring rules for interval-censored data, where Y is unobservable and only (L, R) is observed, we modify the definition of properness:

Definition 4.3 (Proper Scoring Rule for Interval-Censored Data). A scoring rule $\mathcal{S}$ is *proper* if:

$$\mathbb{E}_{L,R}[\mathcal{S}(\hat{F}_Y, (L, R))] \geq \mathbb{E}_{L,R}[\mathcal{S}(F_Y, (L, R))]. \quad (3)$$

We distinguish between two distinct forms of strict properness for interval-censored data:

Definition 4.4 (Strictly Y -proper). A scoring rule is *strictly Y -proper on D* if the equality in (3) implies

$$\hat{F}_Y(y) = F_Y(y) \quad (4)$$

for all $y \in D$.

Definition 4.5 (Strictly LR -proper). A scoring rule is *strictly LR -proper on J* if the equality in (3) implies

$$\hat{F}_Y(r) - \hat{F}_Y(l) = F_Y(r) - F_Y(l) \quad (5)$$

for all $(l, r) \in J$.

We focus on the interval-censored logarithmic score ($\mathcal{S}_{IC-Log}$), which generalizes both the standard log-score (2) and the scoring rule for right-censored data (Rindt et al., 2022):

$$\mathcal{S}_{IC-Log}(\hat{F}_Y, (l, r)) = \begin{cases} -\log(\hat{F}_Y(r) - \hat{F}_Y(l)) & \text{if } l < r, \\ -\log \frac{d}{dt} \hat{F}_Y(l) & \text{if } l = r. \end{cases} \quad (6)$$

Note that this scoring rule without the negative sign is equivalent to the log-likelihood function for the Case-2 interval-censored data (see, e.g., (Huang & Wellner, 1997; Anderson-Bergman, 2017)).

Note also that, while Avati et al. (2020) have utilized this score (6), formal proofs of its properness have been absent. We provide these missing foundations via the following theorems.

Theorem 4.6. *For Case- K interval-censored data, the scoring rule $\mathcal{S}_{IC-Log}$ (6) is strictly Y -proper on D_{LR} under Assumption 3.1.*

Theorem 4.7. *For PIC data, the scoring rule $\mathcal{S}_{IC-Log}$ (6) is strictly Y -proper on D_{PIC} under Assumption 3.1.*

Note that these theorems show strict Y -properness on *subsets* of D_Y (i.e., D_{LR} or D_{PIC}) due to the censoring.

Proofs of these theorems are given in Appendices B.1 and B.2. Note that, while the proof for Theorem 4.6 is a natural generalization of the proof for right-censored data (Yanagisawa, 2023), the proof for Theorem 4.7 is modified to handle the exact observation by using a similar proof strategy used in (Rindt et al., 2022).

Under the weaker assumption of the constant-sum assumption, we show the scoring rule is strictly LR -proper in these two theorems (see Appendices B.3–B.4 for full proofs).

Theorem 4.8. *For Case- K interval-censored data, the scoring rule $\mathcal{S}_{IC-Log}$ (6) is strictly LR -proper on J under Assumption 3.3.*

Theorem 4.9. *For PIC data, the scoring rule $\mathcal{S}_{IC-Log}$ (6) is strictly Y -proper on D_{exact} and strictly LR -proper on J under Assumption 3.3.*

Note that the proof strategy used to prove these theorems are similar to the one on maximum likelihood for interval-censored data (Oller et al., 2004; Piqué, 2006).

Significance of Results in Terms of Survival Analysis. Our framework represents a theoretical advancement over (Rindt et al., 2022). While their proof for survival analysis requires the restrictive independent monitoring assumption (Assumption 3.1), we establish strict properness under the more general constant-sum condition (Assumption 3.3).

Distinction between Strict Y -Properness and Strict LR -Properness. If a scoring rule is strictly LR -proper on J , one might expect that this scoring rule is also strictly Y -proper on D_{LR} . This is usually true, but we show that this can be occasionally false by showing a counter-example: suppose that the underlying distribution of the data is

$$\begin{aligned} z_1 &:= \Pr(L = 0, R = 2, 0 < Y \leq 1) = 1/6, \\ z_2 &:= \Pr(L = 0, R = 2, 1 < Y \leq 2) = 1/6, \\ z_3 &:= \Pr(L = 1, R = 3, 1 < Y \leq 2) = 1/6, \\ z_4 &:= \Pr(L = 1, R = 3, 2 < Y \leq 3) = 1/6, \\ z_5 &:= \Pr(L = 2, R = 4, 2 < Y \leq 3) = 1/6, \\ z_6 &:= \Pr(L = 2, R = 4, 3 < Y \leq 4) = 1/6. \end{aligned}$$

Then the observable interval probabilities are

$$\begin{aligned} \Pr(L = 0, R = 2) &= 1/3, \\ \Pr(L = 1, R = 3) &= 1/3, \\ \Pr(L = 2, R = 4) &= 1/3 \end{aligned}$$

and we have

$$F_Y(0) = 0, F_Y(1) = 1/6, F_Y(2) = 1/2, F_Y(3) = 5/6,$$

and $F_Y(4) = 1$. Then setting

$$\hat{F}_Y(0) = 0, \hat{F}_Y(1) = 1/3, \hat{F}_Y(2) = 1/2, \hat{F}_Y(3) = 1,$$

and $\hat{F}_Y(4) = 1$ satisfies strict LR -properness (5) but violates strict Y -properness (4).

The question of when strict LR -properness implies strict Y -properness hinges on whether the underlying distribution of the data is identifiable from the observed interval probabilities. In the counter-example, the observed probabilities $\Pr(L = 0, R = 2) = 1/3$, $\Pr(L = 1, R = 3) = 1/3$, and $\Pr(L = 2, R = 4) = 1/3$, and the ground-truth probabilities $F_Y(2) - F_Y(0) = 1/2$, $F_Y(3) - F_Y(1) = 2/3$, and

$F_Y(4) - F_Y(2) = 1/2$ imply the linear system

$$\begin{aligned} z_1 + z_2 &= 1/3, \\ z_3 + z_4 &= 1/3, \\ z_5 + z_6 &= 1/3, \\ z_1 + z_2 + z_3 &= 1/2, \\ z_2 + z_3 + z_4 + z_5 &= 2/3, \\ z_4 + z_5 + z_6 &= 1/2. \end{aligned}$$

Writing this as $Az = b$, strict LR -properness guarantees strict Y -properness only if the system has a unique solution (equivalently, A has full column rank). In our counterexample, A is rank-deficient, so multiple underlying distributions are compatible with the same observed interval probabilities; therefore strict LR -properness does not imply strict Y -properness.

5. Calibration Metric

In probabilistic forecasting, calibration assesses the correspondence between a model's predicted probabilities and the observed empirical frequencies (Guo et al., 2017). For a predictive estimator $\hat{F}_Y(y|\mathbf{x})$, achieving calibration is essential for the reliability and interpretability of the predicted CDFs in downstream decision-making.

5.1. Background: Probability Integral Transform

Following (Kuleshov et al., 2018) and (Zhao et al., 2020), we review the standard definition of (average) calibration for fully observed targets through the lens of the Probability Integral Transform (PIT) (David & Johnson, 1948; Angus, 1994).

Definition 5.1 (Standard Calibration (Kuleshov et al., 2018)). A predicted CDF $\hat{F}_Y(y|\mathbf{x})$ is calibrated if the PIT follows a uniform distribution:

$$\Pr_{\mathbf{X}, Y}(\hat{F}_Y(Y|\mathbf{X}) \leq u) = u, \quad \forall u \in [0, 1].$$

To operationalize this as a quantitative metric, we introduce the auxiliary indicator function

$$\pi(u, \alpha) = \begin{cases} 0 & \text{if } u < \alpha, \\ 1 & \text{if } \alpha \leq u. \end{cases} \quad (7)$$

The calibration condition is satisfied when $\mathbb{E}_{\mathbf{X}, Y}[\pi(u, \hat{F}_Y(Y|\mathbf{X}))] = u$. This motivates the Predicted Calibration Error (PCE_p) for fully observed data (Dheur & Ben Taieb, 2023):

$$\text{PCE}_p(\hat{F}_Y) = \int_0^1 \left| \mathbb{E}_{\mathbf{X}, Y}[\pi(u, \hat{F}_Y(Y|\mathbf{X}))] - u \right|^p du, \quad (8)$$

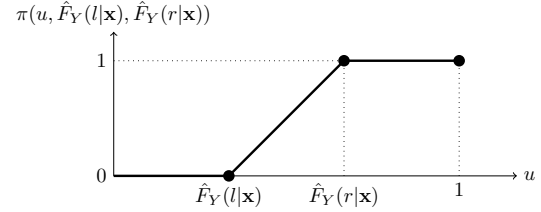


Figure 3. The function $\pi(u, \hat{F}_Y(l|\mathbf{x}), \hat{F}_Y(r|\mathbf{x}))$ is piecewise linear with respect to u .

for $p \geq 1$. This metric is equivalent to the calibration metrics used in (Kuleshov et al., 2018; Utpala & Rai, 2020), and this formulation provides a continuous alternative (Gupta et al., 2021) to discrete binning methods such as expected calibration error (ECE).

5.2. Proposed Metric: IC-Cal

Direct evaluation of Eq. (8) is impossible for interval-censored data, as the exact value of Y is unobservable. We only observe the interval $(L, R]$ such that $L < Y \leq R$. To bridge this gap, we generalize the indicator $\pi(u, \alpha)$ to a bivariate piecewise linear function:

$$\pi(u, \alpha, \beta) = \begin{cases} 0 & \text{if } u < \alpha, \\ \frac{u - \alpha}{\beta - \alpha} & \text{if } \alpha \leq u < \beta, \\ 1 & \text{if } \beta \leq u. \end{cases} \quad (9)$$

As illustrated in Figure 3, π linearly interpolates the ‘‘uncertainty zone’’ created by the censoring interval. When $\alpha = \beta$ (exact observation), Eq. (9) recovers Eq. (7).

Definition 5.2 (Interval-Censored Calibration). An estimator $\hat{F}_Y$ is interval-censored calibrated if:

$$\mathbb{E}_{\mathbf{X}, L, R}[\pi(u, \hat{F}_Y(L|\mathbf{X}), \hat{F}_Y(R|\mathbf{X}))] = u, \quad \forall u \in [0, 1].$$

We propose IC-Cal (γ_{IC}^p) for $p \geq 1$ as a formal metric to quantify the L^p deviation from this identity:

$$\begin{aligned} \gamma_{\text{IC}}^p(\hat{F}_Y) &= \int_0^1 \left| \mathbb{E}_{\mathbf{X}, L, R}[\pi(u, \hat{F}_Y(L|\mathbf{X}), \hat{F}_Y(R|\mathbf{X}))] - u \right|^p du. \end{aligned}$$

The theoretical foundation of IC-Cal is established by the following theorem, which extends the PIT principle to the interval-censored regime.

Theorem 5.3 (Interval-Censored PIT). *Under non-informative censoring (Assumption 3.2), the ground-truth distribution F_Y satisfies:*

$$\mathbb{E}_{\mathbf{X}, L, R}[\pi(u, F_Y(L|\mathbf{X}), F_Y(R|\mathbf{X}))] = u, \quad \forall u \in [0, 1]. \quad (10)$$

The proof of this theorem is given in Appendix B.5.

Given a dataset $\{(\mathbf{x}^{(i)}, l^{(i)}, r^{(i)})\}_{i=1}^N$, the empirical calibration error ($\tilde{\gamma}_{\text{IC}}^p$) of IC-Cal can be computed as

$$\tilde{\gamma}_{\text{IC}}^p(\hat{F}_Y) = \int_0^1 \left| \frac{1}{N} \sum_{i=1}^N \pi(u, \lambda_i, \rho_i) - u \right|^p du, \quad (11)$$

where $\lambda_i = \hat{F}_Y(l^{(i)}|\mathbf{x}^{(i)})$ and $\rho_i = \hat{F}_Y(r^{(i)}|\mathbf{x}^{(i)})$. Note that we can compute the exact value of this empirical error, because $\pi(u, \lambda_i, \rho_i)$ defined in Eq. (9) is piecewise linear in u , as illustrated in Figure 3.

Self-Consistency Equation. In the literature, several non-parametric estimators have been proposed for censored regression based on self-consistency equations (SCEs). For example, the Kaplan-Meier estimator (1958) is known as a solution of an SCE for right-censored data (Efron, 1967), and the nonparametric estimators for interval-censored data are proposed based on SCEs (Turnbull, 1976; Li et al., 1997). Here, we recover the SCE proposed by Li et al. (1997):

$$F_Y(y) = \mathbb{E}_{L,R} [\pi(F_Y(y), F_Y(L), F_Y(R))]$$

by replacing u in Eq. (10) with $F_Y(y)$ and by discarding $\mathbf{X}$. While the validity of this SCE is shown under the independent monitoring assumption (i.e., Assumption 3.1) in (Li et al., 1997), Theorem 5.3 implies that this SCE is also valid under the weaker assumption (i.e., Assumption 3.2). This means that the nonparametric estimator proposed in (Li et al., 1997) is valid under the non-informative censoring assumption.

Significance of IC-Cal. Our calibration metric for interval-censored data generalizes the previously proposed calibration metric for right-censored data (Park et al., 2025), recovering it as special case when $p = 2$. Importantly, our calibration metric is not a generalization of commonly used binning-based metrics such as the expected calibration error (ECE) (Naeini et al., 2015; Guo et al., 2017) or D-calibration (Haider et al., 2020). These binning-based methods are known to be sensitive to bin choice and suffer from interpretability issues (Arrieta-Ibarra et al., 2022; Roelofs et al., 2022). Our formulation avoids binning by using continuous, distribution-free transformations and exact expectations, resulting in a more stable and theoretically grounded calibration framework.

5.3. Theoretical Guarantees

To assess the practical utility of the empirical calibration error $\tilde{\gamma}_{\text{IC}}^p$ (11), we provide a uniform concentration bound. This ensures that a model minimizing the empirical calibration error will converge to the true calibration error as the sample size $N \rightarrow \infty$.

Proposition 5.4 (Concentration of calibration metric).

Let $\delta \in (0, 1)$ and $\mathcal{H}_\pi := \{h : (\mathbf{x}, l, r) \mapsto \pi(u, \hat{F}_Y(l|\mathbf{x}), \hat{F}_Y(r|\mathbf{x})) \mid \hat{F}_Y : \text{estimator}, u \in [0, 1]\}$. With probability at least $1 - \delta$, it holds that

$$\sup_{\hat{F}_Y} \left| \gamma_{\text{IC}}^p(\hat{F}_Y) - \tilde{\gamma}_{\text{IC}}^p(\hat{F}_Y) \right| \leq p \left(2\text{Rad}(\mathcal{H}_\pi) + \sqrt{\frac{2 \log \frac{1}{\delta}}{N}} \right),$$

where $\text{Rad}(\mathcal{H}_\pi)$ is Rademacher complexity of $\mathcal{H}_\pi$, defined by

$$\text{Rad}(\mathcal{H}_\pi) := \mathbb{E}_{(\mathbf{x}^{(i)}, l^{(i)}, r^{(i)})_{i=1}^N} \mathbb{E}_{\sigma} \left[\sup_{h \in \mathcal{H}_\pi} \frac{1}{N} \sum_{i=1}^N \sigma_i h(\mathbf{x}^{(i)}, l^{(i)}, r^{(i)}) \right],$$

where $\sigma \sim \text{Unif}(\{\pm 1\}^N)$.

A proof of this proposition is given in Sec. B.6.

This proposition immediately gives the following bound on the γ_{IC}^p of the minimizer of $\tilde{\gamma}_{\text{IC}}^p$.

Corollary 5.5. Let $\hat{F}_Y$ be a minimizer of $\tilde{\gamma}_{\text{IC}}^p$. Then, with probability at least $1 - \delta$, it holds that

$$\gamma_{\text{IC}}^p(\hat{F}_Y) \leq \inf_{F_Y \in \mathcal{F}_Y} \gamma_{\text{IC}}^p(F_Y) + 2p \left(2\text{Rad}(\mathcal{H}_\pi) + \sqrt{\frac{2 \log \frac{1}{\delta}}{N}} \right).$$

This result implies that as long as $\text{Rad}(\mathcal{H}_\pi) = O\left(\sqrt{\frac{1}{N}}\right)$, the suboptimality gap of $\hat{F}_Y$ in γ_{IC}^p is also bounded by $O\left(\sqrt{\frac{1}{N}}\right)$, which ensures the consistency of the calibration metric-based learning as the sample size goes to infinity.

6. Experiments

In this section, we empirically evaluate the performance of a neural network optimized with the $\mathcal{S}_{\text{IC-Log}}$ scoring rule (6) against established statistical baselines.

6.1. Experimental Setup

NN-IC-Log Model. To assess the efficacy of the proposed scoring rule, we implemented a neural network model, denoted as NN-IC-Log. The architecture consists of a three-layer Multi-Layer Perceptron (MLP) that maps input covariates to a discrete probability distribution. Specifically, the network contains two hidden layers of 32 neurons each, followed by a final output layer. To facilitate robust feature

Table 1. Summary statistics of real-world datasets used in our experiments.

Dataset	Type	Left-censored	Interval-censored	Right-censored	Exact	N
aidsCohort (Y)	PIC	42 (16.3%)	145 (56.4%)	69 (26.8%)	1 (0.4%)	257
aidsCohort (Z)	PIC	1 (0.5%)	22 (11.7%)	147 (78.2%)	18 (9.6%)	188
breastCancer	Case- K	5 (5.3%)	51 (54.3%)	38 (40.4%)	0 (0.0%)	94
hiv	Case- K	25 (6.8%)	78 (21.2%)	265 (72.0%)	0 (0.0%)	368
mastitis	Case- K	26 (6.5%)	291 (72.8%)	83 (20.8%)	0 (0.0%)	400
yoghurt	Case- K	8 (8.6%)	51 (54.8%)	34 (36.6%)	0 (0.0%)	93

representation and mitigate overfitting, we incorporated a Dropout layer (rate = 0.5) and ReLU activation functions after each hidden layer. The output layer utilizes a Softmax function to generate the probability mass from which the CDF is constructed. The model was trained using the Adam optimizer with a learning rate of 1.0, minimizing the $\mathcal{S}_{\text{IC-Log}}$ loss function (6).

Baselines. We compared NN-IC-Log against three representative models from the R statistical ecosystem: **ic_par**, a parametric proportional hazards model provided in the `icenReg` package (Anderson-Bergman, 2017); **ic_sp**, a semi-parametric model also from `icenReg`; and **ICcforest**, a conditional random forest variant tailored for interval-censored data (Yao et al., 2019). In addition, we compared our NN-IC-Log against the neural network model **NN-IC** based on the proportional hazard assumption (Sun & Ding, 2023).

6.2. Datasets and Preprocessing

The models were evaluated on five benchmark datasets from the `icensBKL` package (Bogaerts et al., 2017) in R: `aidsCohort`, `breastCancer` (Beadle et al., 1984a;b), `hiv` (Sun, 2006), `mastitis` (Goethals et al., 2009), and `yoghurt` (Hough, 2010). The dataset characteristics are summarized in Table 1. For the `aidsCohort` dataset, which features dual target variables Y and Z , we excluded 69 observations for variable Z where both interval boundaries were missing to maintain data integrity.

6.3. Results and Discussion

Predictive Performance. The predictive performance, quantified via the $\mathcal{S}_{\text{IC-Log}}$ score, is illustrated in Figure 4. NN-IC-Log demonstrates competitive or superior performance across all scenarios. Notably, NN-IC-Log outperformed ICcforest on the `breastCancer` and `mastitis` datasets. Compared to the parametric `ic_par` baseline, NN-IC-Log exhibited superior performance on `aidsCohort` (target Z), `hiv`, `mastitis`, and `yoghurt`. Our NN-IC-Log showed better performance than NN-IC on `aidsCohort` (target Y) and `hiv`. While the semi-parametric `ic_sp` performed strongly on all the datasets, especially on `aidsCohort` (target Z), NN-

IC-Log showed comparable performance with this model, highlighting its flexibility in capturing complex covariate dependencies.

Calibration Performance. The calibration results, measured using our proposed empirical metric $\tilde{\gamma}_{\text{IC}}^p$ (11) with $p = 2$, are presented in Figure 5. For context, we included two nonparametric estimators (Turnbull, 1976; Li et al., 1997), which are calibrated by design. These results underscore the diagnostic utility of the IC-Cal metric; for instance, the significant miscalibration observed in the `ic_par` model for `aidsCohort` (target Z), `hiv`, and `mastitis` suggests that its parametric assumptions are likely violated in these domains and the significant miscalibration observed in the NN-IC model for `aidsCohort` (target Y), `hiv`, and `mastitis` also suggests some misspecification of the model, whereas our NN-IC-Log remains more closely aligned with the empirical frequencies.

Code Availability. Our Python implementations are available at <https://github.com/CyberAgentAILab/cenreg>.

7. Related Work

Interval-censored data analysis (Huang & Wellner, 1997) has a long history in statistics; classical approaches typically focus on estimating the distribution function $F_Y(y)$ under various monitoring and censoring assumptions. Related, but conceptually distinct, is a line of work on *interval targets*, where each training label specifies an interval believed to contain a latent outcome and the goal is to estimate the latent scalar Y itself. For example, Cheng et al. (2023) proposed a neural network model for learning from interval-valued supervision, Bian et al. (2025) studied boosting-based approaches for interval targets, and Pukdee et al. (2025) proposed another neural network model in this setting. While these methods aim to recover a point estimate of Y consistent with the interval supervision, our work addresses the more general task of estimating the entire cumulative distribution function (CDF) of Y . Because our approach models the full distribution, it can also be specialized to their mean-estimation setting, making it broadly applicable.

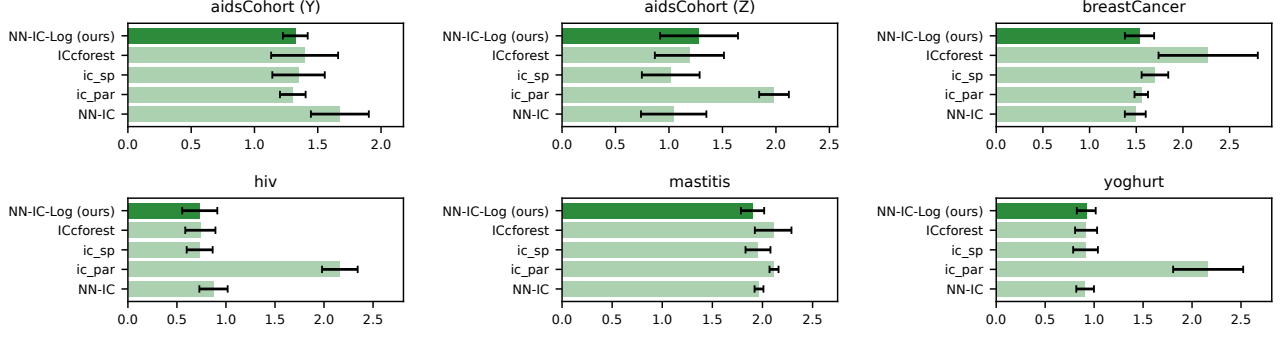


Figure 4. Performance comparison on the scoring rule IC-Log (lower is better) using five-fold cross validation. The error bars correspond to the standard deviation.

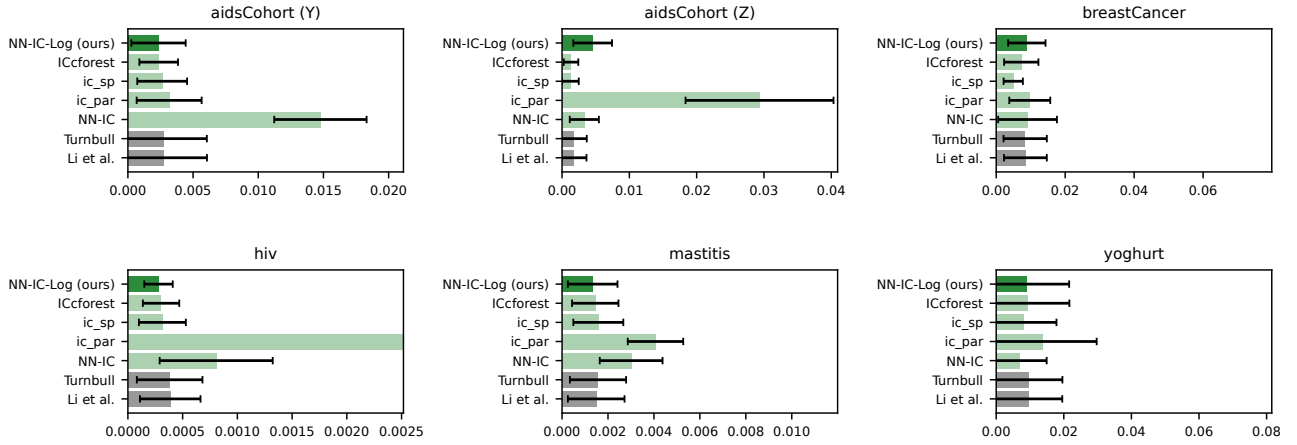


Figure 5. Performance comparison on the empirical calibration error (lower is better) using five-fold cross validation. The error bars correspond to the standard deviation.

8. Conclusion

In this paper, we have formalized the theoretical foundations of scoring rules for interval-censored data by introducing the concepts of strictly Y -proper and strictly LR -proper scoring rules. We established rigorous proofs for the strict Y -properness and LR -properness of the interval-censored logarithmic score under the independent monitoring and constant-sum assumptions, respectively. Our empirical evaluation demonstrates that a neural network optimized with this scoring rule as a loss function performs competitively with, or superior to, specialized statistical baselines. These results are particularly relevant for high-dimensional or non-tabular domains, where traditional statistical methods lack necessary architectural flexibility.

Furthermore, we introduced IC-Cal, a novel calibration metric derived under the non-informative censoring assumption. As a theoretical consequence of our analysis, we demonstrated that the nonparametric estimator previously proposed by Li et al. (1997) remains valid under the weaker non-informative censoring assumption, extending its utility

beyond the strict independent monitoring framework.

A promising direction for future research involves extending the theoretical validity of IC-Cal to the constant-sum assumption—the most general condition in the interval-censoring hierarchy. Such an extension would provide a universal framework for assessing the reliability of probabilistic models across all identifiable censoring mechanisms.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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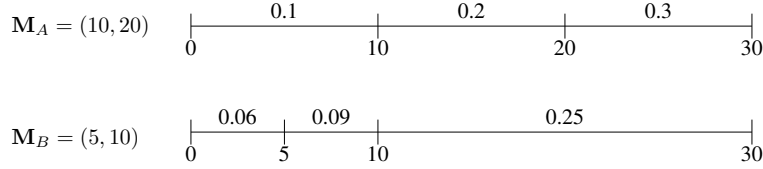


Figure 6. Illustration of an observation scenario that is valid under the assumptions of non-informative censoring but invalid under the stronger assumption of independent monitoring: Monitoring schedules $\mathbf{M}_A = (10, 20)$ and $\mathbf{M}_B = (5, 10)$ are drawn with probabilities 0.6 and 0.4, respectively. The resulting interval observations (L, R) occur with probabilities shown in this figure (e.g., $\Pr(L = 0, R = 10) = \Pr(L = 0, R = 10, \mathbf{M} = (10, 20)) = 0.1$ and $\Pr(L = 0, R = 5) = \Pr(L = 0, R = 5, \mathbf{M} = (5, 10)) = 0.06$).

A. Relationship between Independent Monitoring and Non-Informative Censoring

A.1. Case-2 Interval-Censored Data

The synthetic Case-2 interval-censored data illustrated in Figure 6 demonstrates that non-informative censoring (Assumption 3.2) can be satisfied even when independent monitoring (Assumption 3.1) is explicitly violated. In this figure, two possible monitoring schedules be $\mathbf{M}_A = (10, 20)$ and $\mathbf{M}_B = (5, 10)$, with selection probabilities $\Pr(\mathbf{M}_A) = 0.6$ and $\Pr(\mathbf{M}_B) = 0.4$, respectively.

It is easy to calculate the conditional probabilities for the event time Y at the shared monitoring point $t = 10$:

$$\begin{aligned}\Pr(Y \leq 10 \mid \mathbf{M}_A) &= \frac{0.1}{0.1 + 0.2 + 0.3} = \frac{1}{6}, \\ \Pr(Y \leq 10 \mid \mathbf{M}_B) &= \frac{0.06 + 0.09}{0.06 + 0.09 + 0.25} = \frac{3}{8}.\end{aligned}$$

The discrepancy between these conditional probabilities ($\frac{1}{6} \neq \frac{3}{8}$) indicates a statistical dependence between the event time Y and the monitoring process $\mathbf{M}$. Consequently, the independent monitoring assumption (Assumption 3.1) is violated.

Despite this dependence, Assumption 3.2 holds because the observed intervals do not provide additional information regarding the location of Y within those bounds beyond the distributional constraints. Under this non-informative censoring regime, the latent marginal distribution of Y remains identifiable and can be recovered as follows:

$$\begin{aligned}F_Y(5) &= \Pr(Y \leq 5, \mathbf{M} = \mathbf{M}_A) + \Pr(Y \leq 5, \mathbf{M} = \mathbf{M}_B) = 0.1 \times \frac{0.06}{0.06 + 0.09} + 0.06 = 0.1, \\ F_Y(10) &= 0.25, \\ F_Y(20) &= 0.55, \\ F_Y(30) &= 1.\end{aligned}$$

This example highlights the restrictive nature of the independent monitoring assumption, which requires the distribution of Y to be invariant across different monitoring schedules. In practice, such a condition is frequently violated by study designs or environmental factors. Therefore, estimators and metrics—such as our proposed IC-Cal—that rely on the weaker non-informative censoring assumption (Assumption 3.2) offer significantly greater practical utility and robustness than those constrained by independent monitoring.

A.2. Case-1 Interval-Censored Data

In this section, we show that Assumption 3.2 implies Assumption 3.1 for Case-1 interval-censored data, which means that Assumptions 3.1 and 3.2 are equivalent. Given that we are examining Case-1 interval-censored data, the vector $\mathbf{M}$ is of unit length, allowing us to treat $\mathbf{M}$ as a scalar. Let $\mathcal{M} = \{m_1, m_2, \dots, m_{|\mathcal{M}|}\}$ be the support of $\mathbf{M}$ such that $0 < m_1 < m_2 < \dots < m_{|\mathcal{M}|} < \infty$. Note that each $m_i \in \mathcal{M}$ satisfies $\Pr(\mathbf{M} = m_i) > 0$.

Let $a_i = \Pr(Y \leq m_i \mid \mathbf{M} = m_i)$. Then proving that Assumption 3.2 implies Assumption 3.1 is equivalent to proving

$$a_i = F_Y(m_i) = \Pr(Y \leq m_i) \tag{12}$$

for any i under Assumption 3.2, because $\Pr(Y \leq m_i | \mathbf{M} = m_i) = a_i = \Pr(Y \leq m_i)$ implies

$$\begin{aligned}
 & \Pr(Y \leq y | \mathbf{M} = m_i) \\
 &= \Pr(Y \leq y | \mathbf{M} = m_i, Y \leq m_i) \Pr(Y \leq m_i | \mathbf{M} = m_i) + \Pr(Y \leq y | \mathbf{M} = m_i, Y > m_i) \Pr(Y > m_i | \mathbf{M} = m_i) \\
 &= \Pr(Y \leq y | L = 0, R = m_i, L < Y \leq R) \Pr(Y \leq m_i | \mathbf{M} = m_i) \\
 &\quad + \Pr(Y \leq y | L = m_i, R = \infty, L < Y \leq R) \Pr(Y > m_i | \mathbf{M} = m_i) \\
 &= \Pr(Y \leq y | Y \leq m_i) \Pr(Y \leq m_i | \mathbf{M} = m_i) + \Pr(Y \leq y | Y > m_i) \Pr(Y > m_i | \mathbf{M} = m_i) \\
 &= \Pr(Y \leq y | Y \leq m_i) \Pr(Y \leq m_i) + \Pr(Y \leq y | Y > m_i) \Pr(Y > m_i) \\
 &= \Pr(Y \leq y),
 \end{aligned}$$

which is equivalent to Assumption 3.1. Note that we used Assumption 3.2 in the third equality.

In the following, we prove Eq. (12) under Assumption 3.2. If $F_Y(m_i) = 1$, Eq. (12) follows immediately, and therefore we assume that $0 < F_Y(m_i) < 1$. For any Borel set $A_i^- \subset (0, m_i]$, we have

$$\begin{aligned}
 & \Pr(Y \in A_i^- | \mathbf{M} = m_i) \\
 &= \Pr(Y \leq m_i | \mathbf{M} = m_i) \Pr(Y \in A_i^- | \mathbf{M} = m_i, Y \leq m_i) = a_i \Pr(Y \in A_i^- | Y \leq m_i) = \frac{a_i}{F_Y(m_i)} \Pr(Y \in A_i^-), \quad (13)
 \end{aligned}$$

where we use Assumption 3.2 in the second equality. Similarly, for any Borel set $A_i^+ \subset (m_i, \infty)$, we have

$$\begin{aligned}
 & \Pr(Y \in A_i^+ | \mathbf{M} = m_i) \\
 &= \Pr(Y > m_i | \mathbf{M} = m_i) \Pr(Y \in A_i^+ | \mathbf{M} = m_i, Y > m_i) = (1 - a_i) \Pr(Y \in A_i^+ | Y > m_i) = \frac{1 - a_i}{1 - F_Y(m_i)} \Pr(Y \in A_i^+). \quad (14)
 \end{aligned}$$

For a fixed $m_k \in \mathcal{M}$, we can take u and v such that $m_{k-1} < u < m_k < v < m_{k+1}$.

Let $B_- = (u, m_k]$. Then, we have

$$\begin{aligned}
 \Pr(Y \in B_-) &= \sum_i \Pr(\mathbf{M} = m_i) \Pr(Y \in B_- | \mathbf{M} = m_i) \\
 &= \sum_{i \geq k} \Pr(\mathbf{M} = m_i) \Pr(Y \in B_- | \mathbf{M} = m_i) + \sum_{i < k} \Pr(\mathbf{M} = m_i) \Pr(Y \in B_- | \mathbf{M} = m_i) \\
 &= \Pr(Y \in B_-) \left(\sum_{i \geq k} \Pr(\mathbf{M} = m_i) \frac{a_i}{F_Y(m_i)} + \sum_{i < k} \Pr(\mathbf{M} = m_i) \frac{1 - a_i}{1 - F_Y(m_i)} \right),
 \end{aligned}$$

where we use Eq. (13) and (14) in the last equality since $B_- \subset (0, m_i]$ for $m_i \geq m_k$ and $B_- \subset (m_i, \infty)$ for $m_i < m_k$. Since $\Pr(Y \in B_-) > 0$, we obtain

$$1 = \sum_{i \geq k} \Pr(\mathbf{M} = m_i) \frac{a_i}{F_Y(m_i)} + \sum_{i < k} \Pr(\mathbf{M} = m_i) \frac{1 - a_i}{1 - F_Y(m_i)}. \quad (15)$$

Let $B_+ = (m_k, v]$. By using a similar argument, it holds that

$$\begin{aligned}
 \Pr(Y \in B_+) &= \sum_i \Pr(\mathbf{M} = m_i) \Pr(Y \in B_+ | \mathbf{M} = m_i) \\
 &= \sum_{i > k} \Pr(\mathbf{M} = m_i) \Pr(Y \in B_+ | \mathbf{M} = m_i) + \sum_{i \leq k} \Pr(\mathbf{M} = m_i) \Pr(Y \in B_+ | \mathbf{M} = m_i) \\
 &= \Pr(Y \in B_+) \left(\sum_{i > k} \Pr(\mathbf{M} = m_i) \frac{a_i}{F_Y(m_i)} + \sum_{i \leq k} \Pr(\mathbf{M} = m_i) \frac{1 - a_i}{1 - F_Y(m_i)} \right).
 \end{aligned}$$

Since $\Pr(Y \in B_+) > 0$, we obtain

$$1 = \sum_{i>k} \Pr(\mathbf{M} = m_i) \frac{a_i}{F_Y(m_i)} + \sum_{i \leq k} \Pr(\mathbf{M} = m_i) \frac{1 - a_i}{1 - F_Y(m_i)}. \quad (16)$$

By subtracting Eq. (16) from Eq. (15), we obtain

$$0 = \Pr(\mathbf{M} = m_k) \frac{a_k}{F_Y(m_k)} - \Pr(\mathbf{M} = m_k) \frac{1 - a_k}{1 - F_Y(m_k)}.$$

By rearranging the equality, we get $a_k = F_Y(m_k)$, which is equal to Eq. (12).

B. Proofs

In this section, we present the proofs that were omitted in the main body of this paper.

B.1. Proof of Theorem 4.6

Proof. For any $\mathbf{M} \sim F_{\mathbf{M}}$ such that $\mathbf{M} = (M_1, M_2, \dots, M_K)$, let $p_j^{\mathbf{M}} = \Pr(M_j < Y \leq M_{j+1} | \mathbf{M})$ for $0 \leq j \leq K$. Since Assumption 3.1 holds, we have

$$p_j^{\mathbf{M}} = F_Y(M_{j+1} | \mathbf{M}) - F_Y(M_j | \mathbf{M}) = F_Y(M_{j+1}) - F_Y(M_j).$$

Hence we have

$$\begin{aligned} & \mathbb{E}_{L,R} [\mathcal{S}(\hat{F}_Y, (L, R))] - \mathbb{E}_{L,R} [\mathcal{S}(F_Y, (L, R))] \\ &= \mathbb{E}_{\mathbf{M}} \left[\sum_{j=0}^K p_j^{\mathbf{M}} \left(\log(F_Y(M_{j+1}) - F_Y(M_j)) - \log(\hat{F}_Y(M_{j+1}) - \hat{F}_Y(M_j)) \right) \right] \\ &= \mathbb{E}_{\mathbf{M}} \left[\sum_{j=0}^K p_j^{\mathbf{M}} (\log p_j^{\mathbf{M}} - \log \hat{p}_j^{\mathbf{M}}) \right] \\ &= \mathbb{E}_{\mathbf{M}} [\text{KL}(\mathbf{p}^{\mathbf{M}} || \hat{\mathbf{p}}^{\mathbf{M}})] \\ &\geq 0, \end{aligned}$$

where $\mathbf{p}^{\mathbf{M}} = (p_0^{\mathbf{M}}, p_1^{\mathbf{M}}, \dots, p_K^{\mathbf{M}})$ and $\text{KL}(\cdot || \cdot)$ is the KL divergence. The equality holds if and only if $\mathbf{p}^{\mathbf{M}} = \hat{\mathbf{p}}^{\mathbf{M}}$ for any $\mathbf{M}$, which is equivalent to $F_Y(y) = \hat{F}_Y(y)$ for all $y \in D_{LR}$. $\square$

B.2. Proof of Theorem 4.7

Proof. For any $(\mathbf{M}, \mathbf{E}) \sim (F_{\mathbf{M}}, F_{\mathbf{E}})$ such that $\mathbf{M} = (M_1, M_2, \dots, M_{|\mathbf{M}|})$ and $\mathbf{E} = (E_0, E_1, \dots, E_{|\mathbf{M}|})$, let $p_j^{\mathbf{M}, \mathbf{E}} = \Pr(M_j < Y \leq M_{j+1} | \mathbf{M}, \mathbf{E})$ for $0 \leq j \leq |\mathbf{M}|$. Since Assumption 3.1 holds, we have

$$p_j^{\mathbf{M}, \mathbf{E}} = F_Y(M_{j+1} | \mathbf{M}, \mathbf{E}) - F_Y(M_j | \mathbf{M}, \mathbf{E}) = F_Y(M_{j+1}) - F_Y(M_j),$$

Let $f_Y(y) = \frac{d}{dy} F_Y(y)$ and $q_j^{\mathbf{M}, \mathbf{E}}(y) = f_Y(y) / p_j^{\mathbf{M}, \mathbf{E}}$ for $M_j < y \leq M_{j+1}$, which means that

$$\int_{M_j}^{M_{j+1}} q_j^{\mathbf{M}, \mathbf{E}}(y) dy = \int_{M_j}^{M_{j+1}} \frac{f_Y(y)}{p_j^{\mathbf{M}, \mathbf{E}}} dy = 1.$$

Then we have

$$\begin{aligned}
 & \mathbb{E}_{L,R} [\mathcal{S}(\hat{F}_Y, (L, R))] \\
 &= - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} (1 - E_j) p_j^{\mathbf{M}, \mathbf{E}} \log \hat{p}_j^{\mathbf{M}, \mathbf{E}} \right] - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j \int_{M_j}^{M_{j+1}} f_Y(y) \log \hat{f}_Y(y) dy \right] \\
 &= - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} (1 - E_j) p_j^{\mathbf{M}, \mathbf{E}} \log \hat{p}_j^{\mathbf{M}, \mathbf{E}} \right] \\
 &\quad - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j \int_{M_j}^{M_{j+1}} f_Y(y) \log \frac{\hat{f}_Y(y)}{\hat{p}_j^{\mathbf{M}, \mathbf{E}}} dy \right] + \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j \int_{M_j}^{M_{j+1}} f_Y(y) \log \frac{1}{\hat{p}_j^{\mathbf{M}, \mathbf{E}}} dy \right] \\
 &= - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} (1 - E_j) p_j^{\mathbf{M}, \mathbf{E}} \log \hat{p}_j^{\mathbf{M}, \mathbf{E}} \right] \\
 &\quad - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j p_j^{\mathbf{M}, \mathbf{E}} \int_{M_j}^{M_{j+1}} q_j^{\mathbf{M}, \mathbf{E}}(y) \log \hat{q}_j^{\mathbf{M}, \mathbf{E}}(y) dy \right] - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j p_j^{\mathbf{M}, \mathbf{E}} \log \hat{p}_j^{\mathbf{M}, \mathbf{E}} \right] \\
 &= - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} p_j^{\mathbf{M}, \mathbf{E}} \log \hat{p}_j^{\mathbf{M}, \mathbf{E}} \right] - \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j p_j^{\mathbf{M}, \mathbf{E}} \int_{M_j}^{M_{j+1}} q_j^{\mathbf{M}, \mathbf{E}}(y) \log \hat{q}_j^{\mathbf{M}, \mathbf{E}}(y) dy \right].
 \end{aligned}$$

Hence we have

$$\begin{aligned}
 & \mathbb{E}_{L,R} [\mathcal{S}(\hat{F}_Y, (L, R))] - \mathbb{E}_{L,R} [\mathcal{S}(F_Y, (L, R))] \\
 &= \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} p_j^{\mathbf{M}, \mathbf{E}} (\log p_j^{\mathbf{M}, \mathbf{E}} - \log \hat{p}_j^{\mathbf{M}, \mathbf{E}}) \right] \\
 &\quad + \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\sum_{j=0}^{|\mathbf{M}|} E_j p_j^{\mathbf{M}, \mathbf{E}} \int_{M_j}^{M_{j+1}} q_j^{\mathbf{M}, \mathbf{E}}(y) (\log q_j^{\mathbf{M}, \mathbf{E}}(y) - \log \hat{q}_j^{\mathbf{M}, \mathbf{E}}(y)) dy \right] \\
 &= \mathbb{E}_{\mathbf{M}, \mathbf{E}} \left[\text{KL}(\mathbf{p}^{\mathbf{M}, \mathbf{E}} \| \hat{\mathbf{p}}^{\mathbf{M}, \mathbf{E}}) + \sum_{j=0}^{|\mathbf{M}|} E_j p_j^{\mathbf{M}, \mathbf{E}} \cdot \text{KL}(q_j^{\mathbf{M}, \mathbf{E}} \| \hat{q}_j^{\mathbf{M}, \mathbf{E}}) \right] \\
 &\geq 0,
 \end{aligned}$$

where $\mathbf{p}^{\mathbf{M}, \mathbf{E}} = (p_0^{\mathbf{M}, \mathbf{E}}, p_1^{\mathbf{M}, \mathbf{E}}, \dots, p_{|\mathbf{M}|}^{\mathbf{M}, \mathbf{E}})$ and $\text{KL}(\cdot \| \cdot)$ is the KL divergence. The equality holds if and only if $\mathbf{p}^{\mathbf{M}, \mathbf{E}} = \hat{\mathbf{p}}^{\mathbf{M}, \mathbf{E}}$ and $q_j^{\mathbf{M}, \mathbf{E}} = \hat{q}_j^{\mathbf{M}, \mathbf{E}}$ for any $j \in \{0, 1, \dots, |\mathbf{M}|\}$, $\mathbf{M}$, and $\mathbf{E}$, which is equivalent to $F_Y(y) = \hat{F}_Y(y)$ for any $y \in D_{\text{PIC}}$. $\square$

B.3. Proof of Theorem 4.8

Proof. Let

$$\begin{aligned}
 f_{LR}(l, r) &= \Pr(L = l, R = r) \\
 &= \Pr(l < Y \leq r) \Pr(L = l, R = r | l < Y \leq r) \\
 &= (F_Y(r) - F_Y(l)) k_{LR}(l, r).
 \end{aligned}$$

By definition, we have

$$\sum_{(l,r) \in J} f_{LR}(l, r) = \sum_{(l,r) \in J} (F_Y(r) - F_Y(l)) k_{LR}(l, r) = 1. \quad (17)$$

Analogously we can represent an estimation $\hat{f}_{LR}$ of f_{LR} as $\hat{f}_{LR} = (\hat{F}_Y(r) - \hat{F}_Y(l))k_{LR}(l, r)$ by using an estimation $\hat{F}_Y$ of F_Y . Now we show that

$$\sum_{(l,r) \in J} \hat{f}_{LR}(l, r) = \sum_{(l,r) \in J} (\hat{F}_Y(r) - \hat{F}_Y(l))k_{LR}(l, r) = 1. \quad (18)$$

By using the constant-sum condition, we have

$$\int_y \left(\sum_{(l,r) \in J_y} k_{LR}(l, r) \right) \hat{f}_Y(y) dy = 1.$$

This equation is equivalent to

$$\sum_{(l,r) \in J} (\hat{F}_Y(r) - \hat{F}_Y(l))k_{LR}(l, r) = 1.$$

Therefore $\hat{f}_{LR}(l, r)$ satisfies Eq. (18).

Since Eqs. (17) and (18) hold, we have

$$\begin{aligned} & \mathbb{E}_{L,R} [S(\hat{F}_Y, (L, R))] - \mathbb{E}_{L,R} [S(F_Y, (L, R))] \\ &= - \sum_{(l,r) \in J} f_{LR}(l, r) \log(\hat{F}_Y(r) - \hat{F}_Y(l)) + \sum_{(l,r) \in J} f_{LR}(l, r) \log(F_Y(r) - F_Y(l)) \\ &= \sum_{(l,r) \in J} f_{LR}(l, r) \log \frac{F_Y(r) - F_Y(l)}{\hat{F}_Y(r) - \hat{F}_Y(l)} \\ &= \sum_{(l,r) \in J} f_{LR}(l, r) \log \frac{(F_Y(r) - F_Y(l))k_{LR}(l, r)}{(\hat{F}_Y(r) - \hat{F}_Y(l))k_{LR}(l, r)} \\ &= \sum_{(l,r) \in J} f_{LR}(l, r) \log \frac{f_{LR}(l, r)}{\hat{f}_{LR}(l, r)} \\ &= \text{KL}(f_{LR} || \hat{f}_{LR}) \\ &\geq 0, \end{aligned}$$

where $\text{KL}(\cdot || \cdot)$ is the KL divergence and the equality holds if and only if $f_{LR}(l, r) = \hat{f}_{LR}(l, r)$ holds for any $(l, r) \in J$. This condition is equivalent to $F_Y(r) - F_Y(l) = \hat{F}_Y(r) - \hat{F}_Y(l)$ for any $(l, r) \in J$. $\square$

B.4. Proof of Theorem 4.9

Proof. We consider the joint CDF $F_{\widetilde{LR}}(l, r) = \Pr(\tilde{L} \leq l, \tilde{R} \leq r)$ instead of $F_{LR}(l, r)$. Let $k_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = \Pr(\tilde{L} = \tilde{l}, \tilde{R} = \tilde{r} | \tilde{l} < Y \leq \tilde{r})$ for $(\tilde{l}, \tilde{r}) \in \tilde{J}$, where $\tilde{J} = \{(\tilde{l}, \tilde{r}) | \Pr(\tilde{L} = \tilde{l}, \tilde{R} = \tilde{r}) > 0\}$ be the support of joint probability distribution for $\tilde{L}$ and $\tilde{R}$. By definition, we have $f_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = (F_Y(\tilde{r}) - F_Y(\tilde{l}))k_{\widetilde{LR}}(\tilde{l}, \tilde{r})$, which means that

$$\sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} f_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} (F_Y(\tilde{r}) - F_Y(\tilde{l}))k_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = 1.$$

Now we want to show that a similar equation holds for $\hat{f}_{\widetilde{LR}} = (\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l}))k_{\widetilde{LR}}(\tilde{l}, \tilde{r})$:

$$\sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \hat{f}_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} (\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l}))k_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = 1. \quad (19)$$

Since the constant-sum condition assumes that we have

$$\sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}_y} k_{\widetilde{LR}}(\tilde{l}, \tilde{r}) = 1$$

for any y , where $\tilde{J}_y = \{(\tilde{l}, \tilde{r}) \in \tilde{J} \mid \tilde{l} < y \leq \tilde{r}\}$, we have

$$\int \left(\sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}_y} k_{\tilde{LR}}(\tilde{l}, \tilde{r}) \right) \hat{f}_Y(y) dy = 1.$$

This equation is equivalent to

$$\sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} (\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) k_{\tilde{LR}}(\tilde{l}, \tilde{r}) = 1.$$

Therefore, assuming that $k_{\tilde{LR}}(\tilde{l}, \tilde{r})$ satisfies the constant-sum property, $\hat{f}_{\tilde{LR}}(\tilde{l}, \tilde{r})$ satisfies Eq. (19).

Let $q^{\tilde{l}, \tilde{r}}(y) = f_Y(y) / (F_Y(\tilde{r}) - F_Y(\tilde{l}))$ for $\tilde{l} < y \leq \tilde{r}$, which means that

$$\int_{\tilde{l}}^{\tilde{r}} q^{\tilde{l}, \tilde{r}}(y) dy = \int_{\tilde{l}}^{\tilde{r}} \frac{f_Y(y)}{F_Y(\tilde{r}) - F_Y(\tilde{l})} dy = 1.$$

And let $e(\tilde{l}, \tilde{r}) = 1$ if there exists an interval that satisfy $M_i = \tilde{l}$, $M_{i+1} = \tilde{r}$, and $E_i = 1$ for some $(\mathbf{M}, \mathbf{E}) \sim (F_{\mathbf{M}}, F_{\mathbf{E}})$ and $e(\tilde{l}, \tilde{r}) = 0$ otherwise.

Then we have

$$\begin{aligned} & \mathbb{E}_{L, R} [S(\hat{F}_Y, (L, R))] \\ &= - \sum_{(l, r) \in J} f_{LR}(l, r) \log(\mathbb{1}_{l=r} \cdot \hat{f}_Y(l) + \mathbb{1}_{l \neq r} \cdot (\hat{F}_Y(r) - \hat{F}_Y(l))) \\ &= - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \left(\mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} \frac{f_Y(y)}{F_Y(\tilde{r}) - F_Y(\tilde{l})} \log \hat{f}_Y(y) dy + \mathbb{1}_{e(\tilde{l}, \tilde{r})=0} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \log(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) \right) \\ &= - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} \frac{f_Y(y)}{F_Y(\tilde{r}) - F_Y(\tilde{l})} \log \frac{\hat{f}_Y(y)}{\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})} dy \\ &\quad + \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} \frac{f_Y(y)}{F_Y(\tilde{r}) - F_Y(\tilde{l})} \log \frac{1}{\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})} dy \\ &\quad - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=0} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \log(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) \\ &= - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} q^{\tilde{l}, \tilde{r}}(y) \log \hat{q}^{\tilde{l}, \tilde{r}}(y) dy \\ &\quad - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \log(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=0} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \log(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) \\ &= - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} q^{\tilde{l}, \tilde{r}}(y) \log \hat{q}^{\tilde{l}, \tilde{r}}(y) dy - \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} f_{\tilde{LR}}(\tilde{l}, \tilde{r}) \log(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})) \end{aligned}$$

Hence we have

$$\begin{aligned}
 & \mathbb{E}_{L,R} [S(\hat{F}_Y, (L, R))] - \mathbb{E}_{L,R} [S(F_Y, (L, R))] \\
 &= \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \int_{\tilde{l}}^{\tilde{r}} q^{\tilde{l}, \tilde{r}}(y) \log \frac{q^{\tilde{l}, \tilde{r}}(y)}{\hat{q}^{\tilde{l}, \tilde{r}}(y)} dy + \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \log \frac{F_Y(\tilde{r}) - F_Y(\tilde{l})}{\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})} \\
 &= \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \text{KL}(q^{\tilde{l}, \tilde{r}} \parallel \hat{q}^{\tilde{l}, \tilde{r}}) + \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \log \frac{(F_Y(\tilde{r}) - F_Y(\tilde{l}))k_{\widehat{LR}}(\tilde{l}, \tilde{r})}{(\hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l}))k_{\widehat{LR}}(\tilde{l}, \tilde{r})} \\
 &= \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \text{KL}(q^{\tilde{l}, \tilde{r}} \parallel \hat{q}^{\tilde{l}, \tilde{r}}) + \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \log \frac{f_{\widehat{LR}}(\tilde{l}, \tilde{r})}{\hat{f}_{\widehat{LR}}(\tilde{l}, \tilde{r})} \\
 &= \sum_{(\tilde{l}, \tilde{r}) \in \tilde{J}} \mathbb{1}_{e(\tilde{l}, \tilde{r})=1} \cdot f_{\widehat{LR}}(\tilde{l}, \tilde{r}) \text{KL}(q^{\tilde{l}, \tilde{r}} \parallel \hat{q}^{\tilde{l}, \tilde{r}}) + \text{KL}(f_{\widehat{LR}} \parallel \hat{f}_{\widehat{LR}}) \\
 &\geq 0,
 \end{aligned}$$

where the equality holds if and only if $f_{\widehat{LR}}(\tilde{l}, \tilde{r}) = \hat{f}_{\widehat{LR}}(\tilde{l}, \tilde{r})$ for any $(\tilde{l}, \tilde{r}) \in \tilde{J}$. This condition of the equality is equivalent to $F_Y(\tilde{r}) - F_Y(\tilde{l}) = \hat{F}_Y(\tilde{r}) - \hat{F}_Y(\tilde{l})$ for any $(\tilde{l}, \tilde{r}) \in \tilde{J}$. This is equivalent to $F_Y(y) = \hat{F}_Y(y)$ for any $y \in D_{\text{exact}}$ and $F_Y(r) - F_Y(l) = \hat{F}_Y(r) - \hat{F}_Y(l)$ for any $(l, r) \in J$. $\square$

B.5. Proof of Theorem 5.3

Proof. Let L' and R' denote copies of L and R , respectively. Then, we derive the following:

$$\begin{aligned}
 u &= \Pr(F_Y(Y) \leq u) \\
 &= \mathbb{E}_{L', R'} [\Pr(F_Y(Y) \leq u | L = L', R = R', L \leq Y \leq R)] \\
 &= \mathbb{E}_{L', R'} [\Pr(F_Y(Y) \leq u | L' \leq Y \leq R')],
 \end{aligned}$$

where Assumption 3.2 is utilized in the final equality. Therefore, to complete the proof of this theorem, it remains to demonstrate that:

$$\mathbb{E}_{L', R'} [\Pr(F_Y(y) \leq u | L' \leq Y \leq R')] = \mathbb{E}_{L', R'} [\pi(u, F_Y(L'), F_Y(R'))].$$

To establish this equation, we verify that the following holds for any fixed l and r such that $l \leq r$:

$$\Pr(F_Y(y) \leq u | l \leq Y \leq r) = \pi(u, F_Y(l), F_Y(r)) = \begin{cases} 0 & \text{if } u < F_Y(l), \\ \frac{u - F_Y(l)}{F_Y(r) - F_Y(l)} & \text{if } F_Y(l) \leq u < F_Y(r), \\ 1 & \text{if } F_Y(r) \leq u. \end{cases}$$

We proceed by considering three cases: (i) $u < F_Y(l)$, (ii) $F_Y(l) \leq u < F_Y(r)$, and (iii) $F_Y(r) \leq u$.

Case (i): For $u < F_Y(l)$, we have:

$$\Pr(F_Y(Y) \leq u | l \leq Y \leq r) = \Pr(F_Y(l) \leq F_Y(Y) \leq u | l \leq Y \leq r) = 0,$$

where the first equality arises from the fact that $F_Y(l) \leq F_Y(Y)$ due to $l \leq Y$, and the second equality follows from the condition $u < F_Y(l)$.

Case (ii): For $F_Y(l) \leq u < F_Y(r)$, we have:

$$\Pr(F_Y(Y) \leq u | l \leq Y \leq r) = \Pr(F_Y(l) \leq F_Y(Y) \leq u | l \leq Y \leq r) = \frac{u - F_Y(l)}{F_Y(r) - F_Y(l)},$$

where the first equality is due to $F_Y(l) \leq F_Y(Y)$ given $l \leq Y$.

Case (iii): For $F_Y(r) \leq u$, $F_Y(Y) \leq F_Y(r) \leq u$ always holds because $Y \leq r$. Hence, we have:

$$\Pr(F_Y(Y) \leq u | l \leq Y \leq r) = 1.$$

By combining the results from Cases (i) through (iii), we arrive at the desired conclusion, thereby completing the proof. $\square$

B.6. Proof of Proposition 5.4

Proof. By Theorem 4.10 of (Wainwright, 2019), for $\delta \in (0, 1)$, we have

$$\sup_{\hat{F}_Y, u} \left| \frac{1}{N} \sum_{i=1}^N \pi(u, \hat{F}_Y(l^{(i)} | \mathbf{x}^{(i)}), \hat{F}_Y(r^{(i)} | \mathbf{x}^{(i)})) - \mathbb{E}_{\mathbf{x}, L, R} [\pi(u, \hat{F}_Y(L | \mathbf{x}), \hat{F}_Y(R | \mathbf{x}))] \right| \leq \Delta := 2\text{Rad}(\mathcal{H}_\pi) + \sqrt{\frac{2 \log \frac{1}{\delta}}{N}},$$

with probability at least $1 - \delta$. Then, for arbitrary $u \in [0, 1]$, we have

$$\begin{aligned} & \left| \mathbb{E}_{\mathbf{x}, L, R} [\pi(u, \hat{F}_Y(L | \mathbf{x}), \hat{F}_Y(R | \mathbf{x}))] - u \right|^p - \left| \frac{1}{N} \sum_{i=1}^N \pi(u, \hat{F}_Y(l^{(i)} | \mathbf{x}^{(i)}), \hat{F}_Y(r^{(i)} | \mathbf{x}^{(i)})) - u \right|^p \\ & \leq p \left| \mathbb{E}_{\mathbf{x}, L, R} [\pi(u, \hat{F}_Y(L | \mathbf{x}), \hat{F}_Y(R | \mathbf{x}))] - \frac{1}{N} \sum_{i=1}^N \pi(u, \hat{F}_Y(l^{(i)} | \mathbf{x}^{(i)}), \hat{F}_Y(r^{(i)} | \mathbf{x}^{(i)})) \right| \leq p\Delta, \end{aligned}$$

where we use $|a|^p - |b|^p \leq p|a - b|$ for $a, b \in [-1, 1]$ for the first inequality. The same bound also holds for

$$\left| \frac{1}{N} \sum_{i=1}^N \pi(u, \hat{F}_Y(l^{(i)} | \mathbf{x}^{(i)}), \hat{F}_Y(r^{(i)} | \mathbf{x}^{(i)})) - u \right|^p - \left| \mathbb{E}_{\mathbf{x}, L, R} [\pi(u, \hat{F}_Y(L | \mathbf{x}), \hat{F}_Y(R | \mathbf{x}))] - u \right|^p,$$

which implies

$$\begin{aligned} \left| \gamma_{\text{IC}}^p(\hat{F}_Y) - \tilde{\gamma}_{\text{IC}}^p(\hat{F}_Y) \right| & \leq \int_0^1 \left| \mathbb{E}_{\mathbf{x}, L, R} [\pi(u, \hat{F}_Y(L | \mathbf{x}), \hat{F}_Y(R | \mathbf{x}))] - u \right|^p \\ & \quad - \left| \frac{1}{N} \sum_{i=1}^N \pi(u, \hat{F}_Y(l^{(i)} | \mathbf{x}^{(i)}), \hat{F}_Y(r^{(i)} | \mathbf{x}^{(i)})) - u \right|^p \mathrm{d}u \\ & \leq \int_0^1 p\Delta \mathrm{d}u = p\Delta. \end{aligned}$$

for arbitrarily $\hat{F}_Y$ with probability at least $1 - \delta$. Thus, we obtain the conclusion. $\square$